

12 — Sail On: Wind 10 m/s @ 90° Shrinks the Cascade

Test 01's vessel + sail enabled, true wind 10 m/s at 90°, vessel 12.5 kn.

The wind → cascade chain (test point B8)

Wind (speed+angle) + vessel speed → apparent wind → sail thrust = 539.93 kW
Sail-adjusted propulsion avg = 11 463 - 540 = 10 923 kW
Cascade uses the ADJUSTED value: $H = 10\,923 \times 5\% = 546.2$ (was 573.15!)
→ $I = 546.2 \rightarrow J = 546.2 \times 0.35 = 191.2 \rightarrow L = 355.0$
Hotel unchanged: 76 → 3.8 / 72.2 (wind doesn't shake the galley)
Tiles: PS = 191.2+3.8 = 195 · SR = 355+72.2 = 427.2

Power Demands (new column: SAIL REDUCTION)

ME would carry 11 463+355 = **11 818**; sail takes **-540** → net **11 278**. ME total = 11 278+3 250 = **14 528**
(header) · Energy 14 528.1×5 000 = **72 640 366**. IL1 Optimization Details shows: "Sail Contribution: 11 463 → 10 923 kW · Sail Power 540 kW (4.7 %)".

Baseline & IL1

All combo FOCs drop (lighter ME): 2.5631 ... 2.6072; baseline (3rd-from-worst) 2.5673 → **12 836.6 t/yr**; IL1 2.5631 → **12 815.4**; savings 21.2.

Battery Benefit — and why it FELL

Canonical backend values at \$780: **165.5 t/yr = \$129 105**. The sail and the battery compete for the same swing: the covered band shrank (195 vs 204.4), so the battery's relative value dropped from 173.7 — even though the ship as a whole burns much less (baseline 12 837 vs 13 307). Mirror image of test 16, where heavier sea inflates both.

⚠ On the live screen this test displayed **\$132 416** = benefit × **800** while IL1 costs used 780 — two responses mixed on one screen. Cause: the profile-restore double-fire (finding #5) let an early calculation run with the form DEFAULT fuel price (constant `DEFAULT_VALUES.FUEL_PRICE = 800`, which also explains the "\$800/ton" Financial badge — finding #4). The backend math itself is consistent; the display race is the bug.

Takeaway: sail power feeds the cascade (sail-adjusted propulsion), shrinking both what the battery covers and what the gensets reserve.